Lattice Point Counting via Commutative Algebra

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Semigroups from Linear Algebra

Let A be a matrix with rational or integral entries. After clearing denominators, every rational matrix relevant to lattice-point questions may be replaced by an integral matrix.

Definition 1.1 (Semigroup)**.** A *semigroup* is a set S equipped with an associative binary operation. In the context of this lecture, we primarily consider additive subsemigroups of $\mathbb{Z}^n$, which are subsets $S \subseteq \mathbb{Z}^n$ that are closed under addition. That is, if $x, y \in S$, then $x + y \in S$.

The basic objects in the lecture are subsets of $\mathbb{Z}^n$ cut out by linear equalities and inequalities. For example, the set of solutions

$$\{x \in \mathbb{Z}^n \mid Ax = 0\}$$

forms an additive semigroup. Similarly, the set of nonnegative solutions

$$\{x \in \mathbb{Z}^n \mid Ax \geq 0\}$$

is a semigroup.

Definition 1.2 (Integral span)**.** Let $v_1, \ldots, v_\ell \in \mathbb{Z}^n$. The integral span of these vectors is

$$\operatorname{span}_{\mathbb{Z}}\{v_1, \ldots, v_\ell\} = \left\{ \sum_{i=1}^{\ell} \lambda_i v_i \,\middle|\, \lambda_i \in \mathbb{Z} \right\}.$$

Definition 1.3 (Positive span)**.** Let $v_1, \ldots, v_\ell \in \mathbb{Z}^n$. The positive span, or rational polyhedral cone generated by these vectors, is

$$\operatorname{pos}\{v_1, \ldots, v_\ell\} = \left\{ \sum_{i=1}^{\ell} \lambda_i v_i \,\middle|\, \lambda_i \geq 0 \right\}.$$

Example 1.4. The set

$$\{x \in \mathbb{Z}^n \,|\, Ax = 0\}$$

is an integral span, and

$$\{x \in \mathbb{Z}^n \,|\, Ax \geq 0\}$$

is the lattice points of a positive span.

Affine Semigroups

Definition 1.5 (Affine semigroup)**.** An *affine semigroup* is a finitely generated subsemigroup

$$S \subseteq (\mathbb{Z}^n, +).$$

That is, we can write

$$S = \mathbb{N}\{v_1, \ldots, v_\ell\} := \left\{ \sum_{i=1}^{\ell} n_i v_i \,\middle|\, n_i \in \mathbb{N} \right\}.$$

Here $v_i \in \mathbb{Z}^n$. Throughout these notes, $\Bbbk$ denotes an algebraically closed field of characteristic 0.

Definition 1.6 (Group generated by a semigroup)**.** Let S be a semigroup. The group generated by S is

$$\mathbb{Z}S = \{s_1 - s_2 \mid s_1, s_2 \in S\}.$$

If $S = \mathbb{N}\{v_1, \dots, v_\ell\}$, then $\mathbb{Z}S = \operatorname{span}_{\mathbb{Z}}\{v_1, \dots, v_\ell\}$.

Definition 1.7 (Rank)**.** The rank of an affine semigroup S is the rank of the abelian group $\mathbb{Z}S$:

$$\operatorname{rank}(S) = \operatorname{rank}_{\mathbb{Z}}(\mathbb{Z}S).$$

If $\mathbb{Z}S = \mathbb{Z}^n$, then $\operatorname{rank}(S) = n$.

Definition 1.8 (Saturation)**.** Let S be an affine semigroup. We say that S is *saturated* if, whenever $x \in \mathbb{Z}S$ and $k \in \mathbb{N}_{>0}$ satisfy

$$kx \in S,$$

then $x \in S$. Equivalently, S is saturated if

$$S = \{x \in \mathbb{Z}S \mid kx \in S \text{ for some } k \in \mathbb{N}_{>0}\}.$$

Remark 1.9. Informally, a saturated semigroup has no missing lattice points inside its rational cone.

Numerical Semigroups and the Frobenius Problem

Definition 1.10 (Numerical semigroup)**.** A numerical semigroup is a subsemigroup $S \subseteq \mathbb{N}$ such that $\mathbb{N} \setminus S$ is finite.

Example 1.11. Let

$$S = \mathbb{N}\{a, b\} = \{ma + nb \mid m, n \in \mathbb{N}\} \subseteq \mathbb{N}.$$

Assume $\gcd(a, b) = 1$. If $\gcd(a, b) \neq 1$, then S generates a proper subgroup of $\mathbb{Z}$, so infinitely many natural numbers are missing from S. Conversely, if $\gcd(a, b) = 1$, the classical Frobenius theorem implies that all sufficiently large natural numbers lie in S.

Theorem 1.12 (Sylvester's theorem)**.** *Let $a, b \in \mathbb{N}_{>0}$ satisfy $\gcd(a, b) = 1$, and let $S = \mathbb{N}\{a, b\}$. Define $f(a, b) = (a - 1)(b - 1)$.*

(1) *If $N \geq f(a, b)$, then*

$$N \in S.$$

The largest integer not in S is $f(a, b) - 1 = ab - a - b$.

(2) *The number of gaps in*

$$\{0, 1, 2, \dots, f(a, b)\}$$

is exactly $\frac{f(a,b)}{2}$.

Equivalently, the Frobenius number of S is

$$F(a, b) = ab - a - b.$$

Example 1.13. Let

$$S = \mathbb{N}\{3, 5\} \subseteq \mathbb{N}.$$

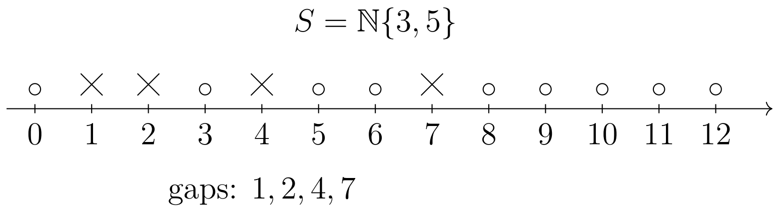
We can compute the elements of S:

$$S = \{0, 3, 5, 6, 8, 9, 10, 11, 12, \ldots\}.$$

The set of gaps is:

$$\mathbb{N} \setminus S = \{1, 2, 4, 7\}.$$

The Frobenius number is $F(3, 5) = 3 \cdot 5 - 3 - 5 = 7$.



Question 1.14 (Three-generator Frobenius problem)**.** Let

$$S = \mathbb{N}\{a, b, c\} \subseteq \mathbb{N}$$

where $\gcd(a, b, c) = 1$. What is $F(a, b, c)$? For example, consider $S = \mathbb{N}\{9, 10, 11\}$. There is no general simple formula for the Frobenius number in three generators.

Examples and Nonexamples of Saturation

Example 1.15. Let

$$S = \mathbb{N}\{3, 5\} \subseteq \mathbb{N}.$$

Because $\mathbb{N} \setminus S = \{1, 2, 4, 7\}$, S is not saturated inside $\mathbb{N}$.

Example 1.16. Let

$$S = \mathbb{N}\{a, b\} \subseteq \mathbb{N}.$$

If $d = \gcd(a, b) \neq 1$, then $\mathbb{Z}S = d\mathbb{Z}$, so every nonnegative integer not divisible by d is missing from S.

Example 1.17 (A nonsaturated affine semigroup)**.** Let

$$S = \mathbb{N}\{(1, 0), (1, 2), (2, 1)\} \subseteq \mathbb{Z}^2.$$

The cone generated by S is the same cone generated by $(1, 0)$ and $(1, 2)$, namely

$$\mathrm{Cone}(S) = \left\{(x, y) \in \mathbb{R}^2 \mid 0 \leq y \leq 2x\right\}.$$

Unlike the semigroup $\mathbb{N}\{(1, 0), (1, 2)\}$, this semigroup generates the whole lattice $\mathbb{Z}^2$. Indeed,

$$\det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1,$$

so the two elements $(1,0)$ and $(2,1)$ already generate $\mathbb{Z}^2$ as a group. Hence

$$\mathbb{Z}S = \mathbb{Z}^2.$$

We now show that S is not saturated. Consider the lattice point

$$(1,1) \in \mathbb{Z}S = \mathbb{Z}^2.$$

This point is not in S. Indeed, if

$$(1,1) = a(1,0) + b(1,2) + c(2,1)$$

with $a, b, c \in \mathbb{N}$, then comparing coordinates gives

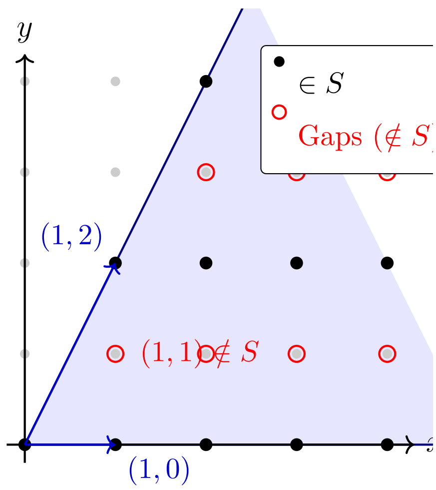
$$a + b + 2c = 1, \qquad 2b + c = 1.$$

The second equation forces either $b = 0, c = 1$, but then the first equation gives $a + 2 = 1$, impossible. Hence $(1,1) \notin S$.

However,

$$2(1,1) = (2,2) = (1,0) + (1,2) \in S.$$

Thus there exists an element $x = (1,1) \in \mathbb{Z}S$ such that $x \notin S$, but $2x \in S$. Therefore S is not saturated.



Pointed Cones and Gordan's Lemma

A central source of affine semigroups is given by lattice points inside rational polyhedral cones.

Definition 1.18 (Pointed cone)**.** Let σ be a cone. We say that σ is *pointed* if

$$\sigma \cap (-\sigma) = \{0\}.$$

Geometrically, this says that the cone has its apex at 0.

Example 1.19. Let

$$\sigma = \{x \in \mathbb{R}^n \mid Ax \geq 0\}$$

be a pointed rational polyhedral cone. Then

$$S = \sigma \cap \mathbb{Z}^n$$

is a semigroup.

Proposition 1.20. *Let σ be a rational polyhedral cone, and set*

$$S = \sigma \cap \mathbb{Z}^n.$$

If $\mathbb{Z}S = \mathbb{Z}^n$, then S is saturated.

Proof. Suppose $x \in \mathbb{Z}S$ and $kx \in S$ for some $k \in \mathbb{N}_{>0}$. Since $S \subseteq \sigma$, we have

$$kx \in \sigma.$$

Since σ is a cone, it is closed under multiplication by positive real scalars. Hence

$$x = \frac{1}{k}(kx) \in \sigma.$$

Therefore $x \in \sigma \cap \mathbb{Z}S$. In the case $\mathbb{Z}S = \mathbb{Z}^n$, this is simply

$$x \in \sigma \cap \mathbb{Z}^n = S.$$

□

Theorem 1.21 (Gordan's lemma)**.** *Let*

$$\sigma = \operatorname{pos}\{v_1, \ldots, v_\ell\} \subseteq \mathbb{R}^n$$

be a rational cone, where $v_1, \ldots, v_\ell \in \mathbb{Z}^n$. Then

$$S = \sigma \cap \mathbb{Z}^n$$

is an affine semigroup (it is finitely generated).

Proof. Let $v \in S$. Since $v \in \sigma$, there exist real numbers $\lambda_i \geq 0$ such that

$$v = \sum_{i=1}^{\ell} \lambda_i v_i.$$

Write $\lambda_i = \lfloor \lambda_i \rfloor + \{\lambda_i\}$, where

$$\lfloor \lambda_i \rfloor \in \mathbb{N}, \qquad 0 \leq \{\lambda_i\} < 1.$$

Then

$$v = \sum_{i=1}^{\ell} \lfloor \lambda_i \rfloor v_i + \sum_{i=1}^{\ell} \{\lambda_i\} v_i.$$

Define the fundamental parallelepiped

$$\operatorname{Par}(v_1, \ldots, v_\ell) = \left\{ \sum_{i=1}^{\ell} \mu_i v_i \,\middle|\, 0 \leq \mu_i < 1 \right\}.$$

Since v and $\sum_{i=1}^{\ell} \lfloor \lambda_i \rfloor v_i$ are lattice points, the second summand is also a lattice point. Hence

$$\sum_{i=1}^{\ell} \{\lambda_i\} v_i \in \operatorname{Par}(v_1, \dots, v_\ell) \cap \mathbb{Z}^n.$$

The intersection $\operatorname{Par}(v_1, \dots, v_\ell) \cap \mathbb{Z}^n$ is a bounded discrete set, hence finite. It follows that every element of S is generated by the finite set

$$\{v_1, \dots, v_\ell\} \cup \Big(\operatorname{Par}(v_1, \dots, v_\ell) \cap \mathbb{Z}^n\Big).$$

□

Remark 1.22. The lecture conclusion was:

$$\text{generators of } S \subseteq \{v_1, \dots, v_\ell\} \cup \Big(\operatorname{Par}(v_1, \dots, v_\ell) \cap \mathbb{Z}^n\Big),$$

which proves the set is finitely generated.

Definition 1.23 (Hilbert basis)**.** Let S be a positive affine semigroup. A minimal set of semigroup generators for S is called the Hilbert basis of S, denoted $\operatorname{Hilb}(S)$. For a positive affine semigroup, an element $g \in S \setminus \{0\}$ belongs to $\operatorname{Hilb}(S)$ if and only if it cannot be written as

$$g = v + v'$$

with nonzero $v, v' \in S$.

Semigroup Rings

Let S be an affine semigroup.

Definition 1.24 (Semigroup ring)**.** The semigroup ring of S over $\Bbbk$ is

$$\Bbbk[S] = \Bbbk[t^m \mid m \in S].$$

For $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$, we write

$$t^m = t_1^{m_1} \cdots t_n^{m_n}.$$

The multiplication rule is $t^m t^{m'} = t^{m+m'}$.

Remark 1.25. When $S \subseteq \mathbb{N}^n$, the ring $\Bbbk[S]$ is a subalgebra of the polynomial ring $\Bbbk[t_1, \dots, t_n]$. More generally, if $S \subseteq \mathbb{Z}^n$, then $\Bbbk[S]$ is naturally a subalgebra of the Laurent polynomial ring

$$\Bbbk[t_1^{\pm 1}, \dots, t_n^{\pm 1}].$$

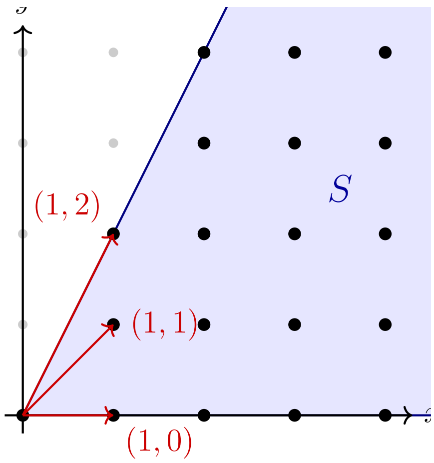
Example 1.26. Let

$$S = \mathbb{N}\{(1,0), (1,1), (1,2)\} \subseteq \mathbb{Z}^2.$$

Then

$$\Bbbk[S] = \Bbbk[t_1, t_1 t_2, t_1 t_2^2] \subseteq \Bbbk[t_1, t_2].$$

Notice that $(1,1) = \frac{1}{2}((1,0) + (1,2)) \in \operatorname{pos}\{(1,0), (1,2)\}$, but $(1,1)$ cannot be written as a sum of other nonzero elements in S, so it is a necessary Hilbert-basis element.



Proposition 1.27. *Let S be an affine semigroup. Then:*

(1) $\Bbbk[S]$ *is an integral domain.*

(2) $\dim \Bbbk[S] = \operatorname{rank}(S)$. *If $S = \sigma \cap \mathbb{Z}^n$, then* $\dim \Bbbk[S] = \dim \sigma$.

(3) $\Bbbk[S]$ *is a finitely generated* $\Bbbk$*-algebra.*

(4) *If S is saturated, then $\Bbbk[S]$ is integrally closed.*

Proof. Since $\Bbbk[S]$ embeds into the Laurent polynomial ring

$$\Bbbk[t_1^{\pm 1}, \dots, t_n^{\pm 1}],$$

it is an integral domain.

The dimension statement follows because the fraction field of $\Bbbk[S]$ is generated by the characters corresponding to the group $\mathbb{Z}S$. Therefore the transcendence degree of $\operatorname{Frac}(\Bbbk[S])$ over $\Bbbk$ is $\operatorname{rank}(S)$, and hence

$$\dim \Bbbk[S] = \operatorname{rank}(S).$$

If $S = \mathbb{N}\{m_1, \dots, m_r\}$, then

$$\Bbbk[S] = \Bbbk[t^{m_1}, \dots, t^{m_r}],$$

which proves it is finitely generated.

Finally, the equivalence between saturation of S and normality of $\Bbbk[S]$ is the standard normality criterion for affine semigroup rings. Thus, if S is saturated, then $\Bbbk[S]$ is integrally closed. □

Remark 1.28. The last point is the algebraic heart of the toric dictionary:

$$S \text{ saturated} \quad \Longleftrightarrow \quad \Bbbk[S] \text{ normal}$$

Affine toric varieties are exactly spectra of saturated affine semigroup rings.

Presentations by Binomial Ideals

Let

$$S = \mathbb{N}\{m_1, \ldots, m_r\} \subseteq \mathbb{Z}^n.$$

There is a surjective $\Bbbk$-algebra homomorphism

$$\varphi : \Bbbk[T_1, \ldots, T_r] \longrightarrow \Bbbk[S]$$

given by

$$T_i \longmapsto t^{m_i}.$$

By the First Isomorphism Theorem,

$$\Bbbk[S] \cong \frac{\Bbbk[T_1, \ldots, T_r]}{\ker \varphi}.$$

Example 1.29. Let

$$S = \mathbb{N}\{(1,0),(1,1),(1,2)\}.$$

Let

$$m_1 = (1,0), \qquad m_2 = (1,1), \qquad m_3 = (1,2).$$

The map is

$$\varphi : \Bbbk[T_1, T_2, T_3] \longrightarrow \Bbbk[S]$$

with

$$T_1 \longmapsto t_1, \qquad T_2 \longmapsto t_1 t_2, \qquad T_3 \longmapsto t_1 t_2^2.$$

Notice that

$$m_1 + m_3 = 2m_2,$$

which implies that

$$\varphi(T_1 T_3) = t_1^2 t_2^2 = \varphi(T_2^2).$$

Thus,

$$T_1 T_3 - T_2^2 \in \ker \varphi.$$

In fact, it is true that $\ker \varphi = (T_1 T_3 - T_2^2)$, so

$$\Bbbk[S] \cong \frac{\Bbbk[T_1, T_2, T_3]}{(T_1 T_3 - T_2^2)}.$$

$$\begin{tikzcd}
\Bbbk[T_1, T_2, T_3] \arrow[r, "\varphi"] \arrow[d, two heads] & \Bbbk[S] \\
\Bbbk[T_1, T_2, T_3]/(T_1 T_3 - T_2^2) \arrow[ur, "\sim"'] &
\end{tikzcd}$$

Theorem 1.30 (Affine semigroup rings have binomial presentations)**.** *Let S be an affine semigroup generated by $m_1, \ldots, m_r$. Then*

$$\Bbbk[S] \cong \frac{\Bbbk[T_1, \ldots, T_r]}{I_S},$$

where I_S is a binomial ideal.

Proof. The map

$$\varphi : \Bbbk[T_1, \dots, T_r] \to \Bbbk[S], \qquad T_i \mapsto t^{m_i},$$

maps monomials to monomials. For $\alpha = (\alpha_1, \dots, \alpha_r) \in \mathbb{N}^r$, we have

$$\varphi(T^\alpha) = t^{\alpha_1 m_1 + \cdots + \alpha_r m_r}.$$

Thus $\varphi(T^\alpha) = \varphi(T^\beta)$ when

$$\sum_{i=1}^r \alpha_i m_i = \sum_{i=1}^r \beta_i m_i.$$

The kernel is generated by the binomials $T^\alpha - T^\beta$. □

Multigrading and Affine Toric Geometry

The semigroup ring $\Bbbk[S]$ is naturally graded by the group $\mathbb{Z}S$:

$$\deg(t^m) = m.$$

Under the presentation

$$\Bbbk[S] \cong \Bbbk[T_1, \dots, T_r]/I_S,$$

this corresponds to setting

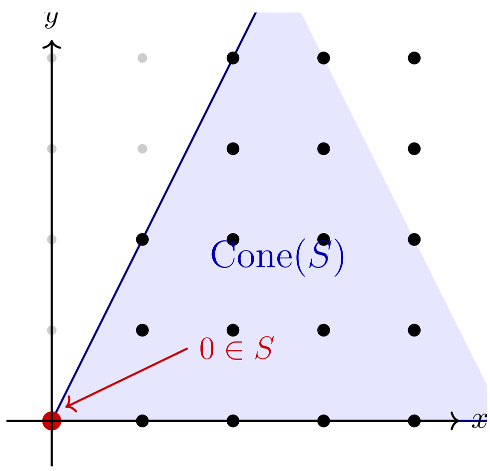
$$\deg(T_i) = m_i.$$

Geometrically, the presentation

$$\Bbbk[S] \cong \frac{\Bbbk[T_1, \dots, T_r]}{I_S}$$

gives a closed embedding of the affine toric variety into affine space:

$$\operatorname{Spec} \Bbbk[S] \hookrightarrow \operatorname{Spec} \Bbbk[T_1, \dots, T_r] \cong \mathbb{A}^r.$$



Remark 1.31. The final note from the lecture is that the minimal graded free resolution of binomial ideals is far from understood. Thus, even though affine semigroup rings have explicit binomial presentations, the homological algebra of their defining ideals can be highly nontrivial.